

Week 19: Connecting Frontend & Backend

1. Making HTTP Requests with Axios (React + Node)

Axios is a popular library used to send HTTP requests between React frontend and Node.js backend. It supports GET, POST, PUT, and DELETE requests. This enables smooth frontend-backend communication.

2. Handling CORS

CORS (Cross-Origin Resource Sharing) allows communication between frontend and backend running on different origins. It prevents browser restrictions and enables secure API access. This is essential in MERN applications.

3. Environment Variables

Environment variables are used to store sensitive or configurable data such as API keys, database URLs, JWT secrets, and port numbers. Files like `.env` keep these values separate from source code. This improves security and flexibility.

4. Error Handling on Both Sides

Proper error handling is important in both frontend and backend. React can display meaningful messages to users, while Node.js handles server-side exceptions. This improves debugging and user experience.

5. API Integration Testing

After connecting frontend and backend, APIs should be tested to verify data flow and response handling. Tools like Postman and frontend integration testing help identify issues quickly.

Conclusion

This week focused on integrating frontend and backend systems. By learning Axios, CORS, environment variables, API testing, and error handling, stronger full-stack development and debugging skills were achieved.